Ashmit Khandelwal

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EDUCATION

Carnegie Mellon University Pittsburgh, PA
M.S. Machine Learning Aug 2026 – Dec 2027
Coursework: Advanced Topics in ML, Probability & Statistics
Research: KV Cache Reuse for Long-Horizon LLM Agents, Prof. Chenyan Xiong

Birla Institute of Technology and Science, Pilani India
B.E. Computer Science, Minor in Data Science. GPA: 3.8/4.0 Nov 2020 - May 2024

EXPERIENCE

Research Fellow, Microsoft Research Jul 2024 – Jun 2026
Advisors: Nagarajan Natarajan, Amit Sharma Bangalore, India

- Developed interwhen, a general test-time verification framework for asynchronously monitoring and steering single LLM trajectories, improving accuracy and policy compliance across code, math, logical, spatial, and agentic benchmarks.
- Formally defined Deep Research (DR), a previously under-defined class of LLM systems for large-scale search and reasoning. Introduced LiveDRBench, one of the first benchmarks to evaluate DR. Evaluated systems from OpenAI, Perplexity, and Gemini, identifying breadth of exploration as a key bottleneck, with the strongest system achieving only 0.55 F1.

Research Intern, Adobe May - Aug 2023
Advisors: Yaman Kumar, Changyou Chen Noida, India

- Modeled human behavior across 40K+ YouTube videos to predict rewatchability, engagement metrics, and audience sentiment at scale across diverse content domains.
- Designed and trained multimodal LLM architectures integrating video and textual representations. Achieved significant cross-domain gains, surpassing GPT-4 on multiple behavior prediction tasks despite using 10× fewer parameters.
- Built pipelines for video, audio, and text scraping and processing, automating collection of terabytes of web data.

PUBLICATIONS

Large Content And Behavior Models To Understand, Simulate, And Optimize Content And Behavior
A. Khandelwal*, A. Agrawal*, A. Bhattacharyya*, Y.K. Singla*, et al. Spotlight at ICLR 2024

Large Content and Behavior Models (LCBMs) reintroduce behavior tokens into LLM training data to simulate and explain human audience behavior in response to different forms of content. These models generalize across content types and adapt to diverse behavior domains, demonstrated on the newly introduced Content Behavior Corpus (CBC).

Characterizing Deep Research: A Benchmark and Formal Definition
A. Java*, A. Khandelwal*, S. Midigeshi*, et al. ICLR 2026

Deep Research (DR) systems lacked a formal definition despite the emergence of several proprietary and open-source models. This work established a concrete definition of DR, introduced LiveDRBench as one of the first DR benchmarks along with a recursive evaluation framework. Highlighted performance gaps in DR systems from OpenAI, Perplexity, and Gemini.

interwhen: A Generalizable Framework for Verifiable Reasoning with Test-time Monitors
V. Bhat*, P. Chanda*, V. Ekbote*, A. Khandelwal, M. Swaroop, et al. NeurIPS 2026

Interwhen is a test-time verification framework for reasoning models to asynchronously verify intermediate reasoning. Improves reasoning soundness across code generation, logical reasoning, and agentic settings, without the high token overhead of standard test-time scaling methods.

Benchmarking VLMs' Reasoning About Persuasive Atypical Images
S. Malakouti*, A. Aghazadeh*, A. Khandelwal, A. Kovashka WACV 2025

Benchmarked vision-language models on inferring the message in rhetorical and atypical imagery used in advertisements through three novel tasks. Found that VLMs struggle with reasoning about atypicality, but simple methods enable atypicality-aware verbalization and improve rhetorical image understanding.

SKILLS

Languages & Frameworks: Python, JavaScript, PyTorch, HuggingFace, vLLM
Research Areas: Large Language Models, Test-Time Scaling, Search, Reasoning, Runtime Verification, Deep Research, Agents, Multimodality, Vision Language Models